
Software Requirements Specification

for

TrailBlazer: Sam Houston Hiking Companion

Version 1.0

Prepared by

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Revision History

Name	Date	Reason For Changes	Version
Falynne Armstrong, Ashley Michallick, Lauren Rosales, Bryan Barber	9/30/2023	First version--basic implementation of Trailblazer to mobile application	1.0

1. Introduction

TrailBlazer: Sam Houston Hiking Companion is a platform that will allow Sam Houston National Forest to provide users an interface to view information on the many hikes available in an easy-to-understand, visual format on mobile application. For the project purpose, TrailBlazer: Sam Houston Hiking Companion will also allow features for people with vision impairments and mobility issues who will be able to acquire the information they need to go on the hike.

1.1 Document Purpose

The software requirement specification for TrailBlazer: Sam Houston Hiking Companion is intended to provide a detailed explanation of the features and design specifications for this application. Additionally, it will serve as a tool to communicate the design plans of TrailBlazer: Sam Houston Hiking Companion team with Dr. Islam so that both parties may fully understand the specifications, requirements, and design of the TrailBlazer: Sam Houston Hiking Companion project.

1.2 Document Conventions

This document uses Time New Roman font throughout the entire document. Major headlines will use bold and underline, size 22 font. Subsections use bold, size 18 font. Smaller subsections use bold, size 14 font. Any part of this document that is not a header of some kind uses size 12 font. The one inch margins are maintained throughout this document as well.

1.3 Intended Audience and Reading Suggestions

This document is intended for faculty and the instructor for COSC 4319, our team, and anyone else interested in learning more about the TrailBlazer: Sam Houston Hiking Companion application.

The remainder of this document provides the general product description and a technical outline for the requirements of this system. Section two will give a high level description of the project functionality and implementation details. Section three will give a more detailed description of the specific requirements for different components of the application, which include various interfaces and functional requirements.

1.4 Product Scope

TrailBlazer: Sam Houston Hiking Companion is a mobile application that is designed to allow users that are new to hiking, visually and handicapped disabilities. This application will give users information about the Sam Houston National Forest. For visual disabilities, our app will provide an audio tour of the hike to help users understand what they can't see. Handicapped accessibility will indicate which trails are accessible to them. Then, for those who are new to hiking, a brief review of the hike and the various equipment required for the hike will be provided. Our system will be available on iOS/Android and will be able to work offline.

1.5 References

- Node.js:
<https://nodejs.org/en>
- React Native:
<https://reactnative.dev/>
- Express.js:
<https://expressjs.com/>
- Html5:
<https://www.w3.org/TR/2011/WD-html5-20110405/>
- CSS:
<https://www.w3.org/Style/CSS/Overview.en.html>
- MySQL:
<https://www.mysql.com/>
- Google Cloud:
<https://cloud.google.com/>

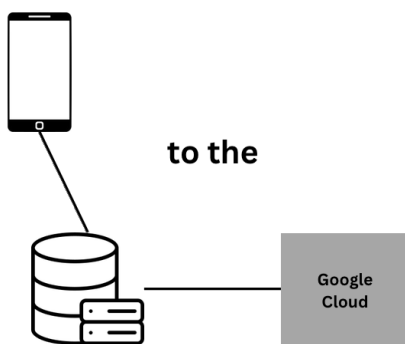
2. Overall Description

2.1 Product Perspective

This is a new, self-contained project. Initially, there is no framework or system in place to build upon; therefore, this software will be built from the ground up. This system will consist of a back-end and front-end. In the front-end, the user will interact directly with the mobile app using the interface. Additionally, the front-end will be used by the users to select which hike they would want to go on. In the back-end, our software engine will allow us to store and pull data on the MySQL. It should be able to manage the connection to the Google Cloud using the API.

TrailBlazer: Sam Houston Hiking Companion will be designed as two parts, the front and back end, that will be used by end users and Sam Houston National Forest respectively. The front end will be a mobile application where the user may view the different trails that are offered at Sam Houston National Forest. The mobile application will be built on top of Node.js, React Native, Express.js, HTML5, and CSS and the interface running partly on Google Cloud with MySQL will store the login information and pull it. Also we will store the achievement on the SQLite to store which hike the user completed.

The back-end will be the SQLite server that will store and push the login with their email and password. The back-end will be primarily used by our group to have access to data. Google Cloud will hold most of the information of the mobile app. When a user installs the app, it connects to a Google Cloud server and downloads all of the app's content. The software will save this data for subsequent use, eliminating the need for constant server connections. only when it is connected to Wi-Fi or a cellular network. The purpose of the API will be to manage this kind of server connection.



2.2 Product Functions

- The user will be able to access to a map of the Sam Houston National Forest
- The user with visually impairments and mobility impairments will have features to provides aid using the app
 - Visual impairment features
 - text-to-speech of the information on the app screen
 - recording of the trails to give the visually impaired user a visualization of the trails
 - Expand of text size
 - Mobility impairments
 - The app will have the information on which trails are wheelchair accessible
- The app is able to be functional without Wi-Fi or data usage
- The user will be able to read and listen to a small summary of each trail
- The app will have a database to hold login information
- The app will have security for the user data is protected by encryption algorithm/ salt

2.3 User Classes and Characteristics

The initial users of our software front end Individuals that are accustomed with navigating a mobile device should be able to glance at the map and view the different paths. The user may take a moment to familiarize himself/herself with the layout and interface of our mobile service, but should easily understand what he/she is looking at.

The users of the back-end, or the SQLite database and the API, will be the TrailBlazer: Sam Houston Hiking Companion's group to have access to the data. Therefore, it is expected that these individuals have a solid foundation with my SQLite and API and are capable of navigating themselves.

2.4 Operating Environment

TrailBlazer: Sam Houston Hiking Companion will primarily act as a mobile app that will be supported by the App Store and Google Play. We assume that the underlying operating system is Apple iOS and Android OS.

TrailBlazer: Sam Houston Hiking Companion will be running on Google Cloud Server that Node.js, React Native, Express.js, HTML5,CSS and MySQL running on it.

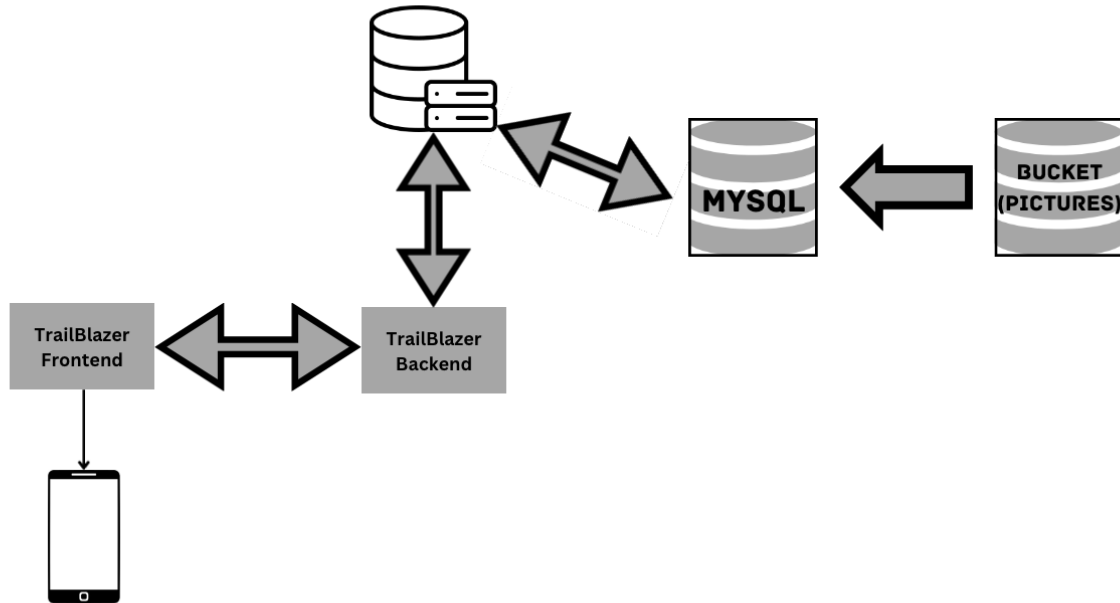


Figure 2. Operating Environment

2.5 Design and Implementation Constraints

Design:

- Language: Node.js, React Native, Express.js, HTML5, CSS3
- Database: MySQL
- Server: Google Cloud

Constraints:

- Must be able to run offline when without internet and cell service
- Can't take too much storage on the phone
- Visual impairment features are needed for this app, such as text-to-speech, etc.
- Wheelchair accessibility information list with the trail

2.6 User Documentation

Everything will be as simple as possible. Ideally, it will require no extra explanation. Every step for account setup and viewing options will be clearly labeled and described as the user progresses through the mobile app.

2.7 Assumptions and Dependencies

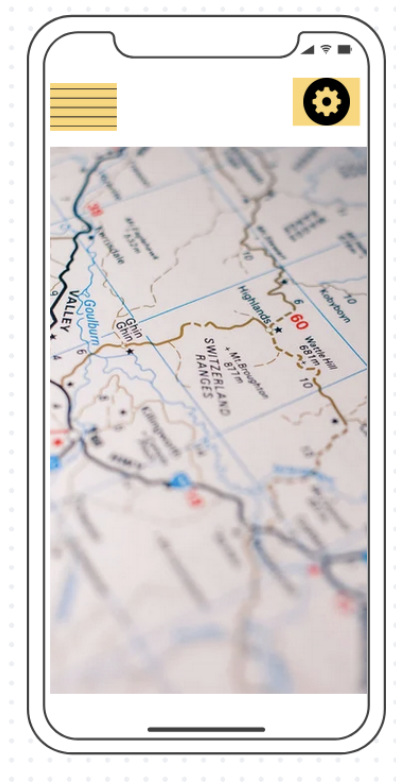
- We assume the user has access to a smartphone
- We assume the user has an internet connection or cellular service in order to download the applications from the app store
- We assume that the user can understand how to use a mobile app
- We depend on the correct operation
- We depend on the server to work when the information is transmitted to the app when it first downloads and connects to the server.

3. External Interface Requirements

3.1 User Interfaces

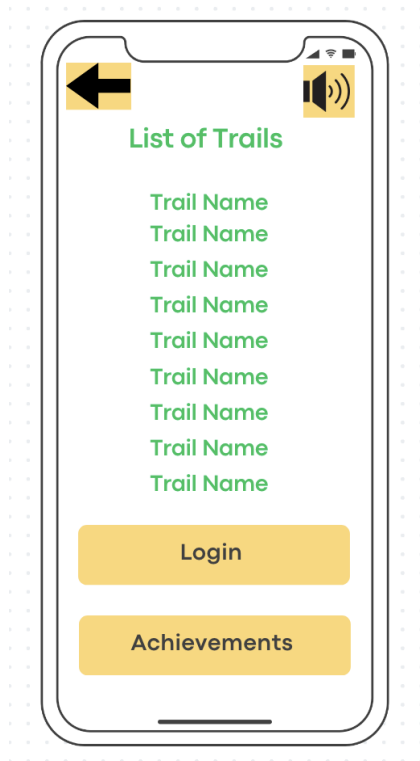
Home screen :

On the Home screen it has the whole map of the Sam Houston National Forest. On the top left of the screen there is a button to take the user to the list of trails on their screen. On the top right of the screen is the setting button it will take you to the setting screen.



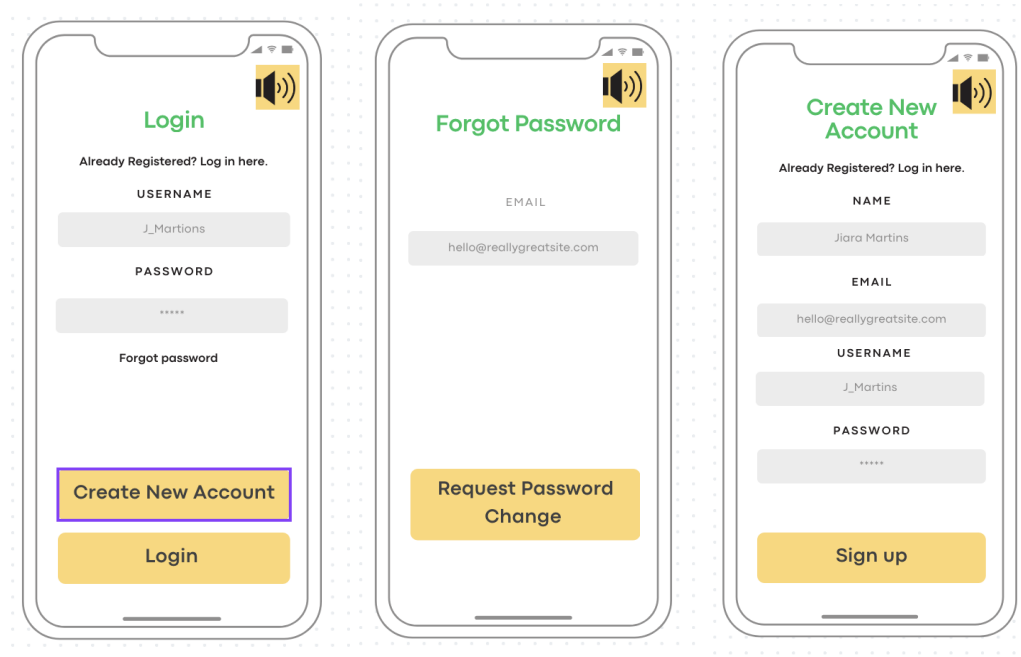
List of trails screen:

In this screen, the application will have all of the hikes offered from Sam Houston National Forest. On the top left of the screen there is a back button. On the top right of the screen is a button to read everything on the screen where it should be. At the bottom of the screen we have the login button to take the user to the login screen, and the achievement button to take the user to the achievement screen.



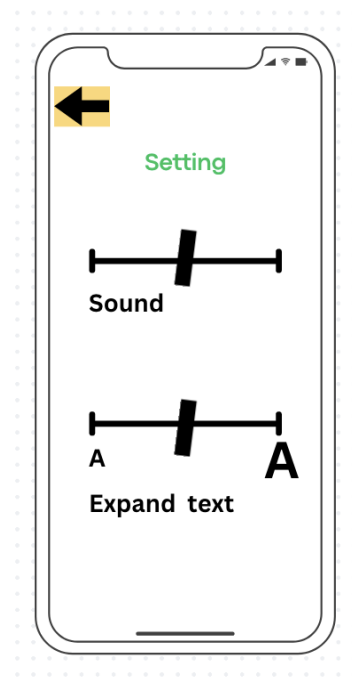
Login screen:

The login screen will allow users to log in to their account using their username and password. The user will then be redirected to the Home screen upon successful login. The create new account screen will provide a new user to create a TrailBlazer: Sam Houston Hiking Companion, so that she or she can still use the app without having to create an account but it will be able to track your achievements. If creating an account the user must enter his or her name, email, username, and password. An error message will be displayed if the user does not enter in one of the fields or if one of the form fields is filled out incorrectly. In the forgot password screen the user puts their email, then an email will be sent to their inbox then they will be able to change their password. Once the user creates a new account by hitting the sign up button, it will redirect it back to the login screen to sign in using your username and password. On the top left of the screen there is a speaking button to read off all of the information on the page.



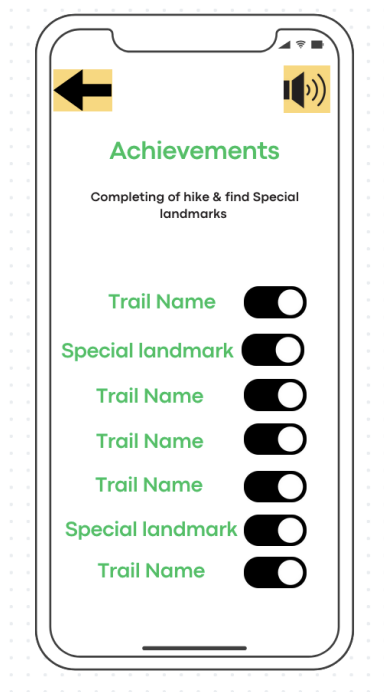
Setting screen:

In this setting screen, the user will be able to control the sound of the app. Also the user will be able to control the setting to expand the text of the app. On the top left of the screen, there is a back button to redirect back to the home screen.



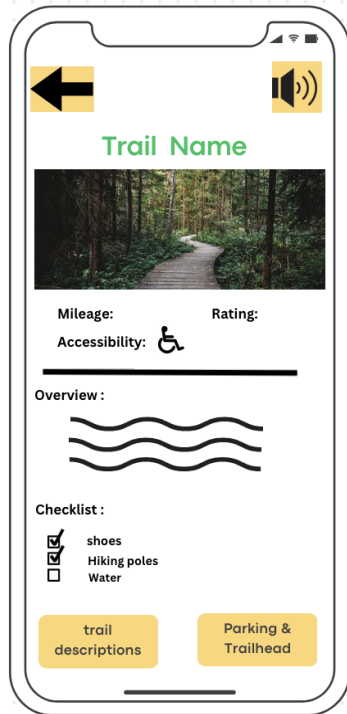
Achievement screen:

In the achievement screen, the user will be able to mark off the different trails the user completes at Sam Houston National Forest. On the top right of the screen, it has a back button to redirect back to the list of trail screens.



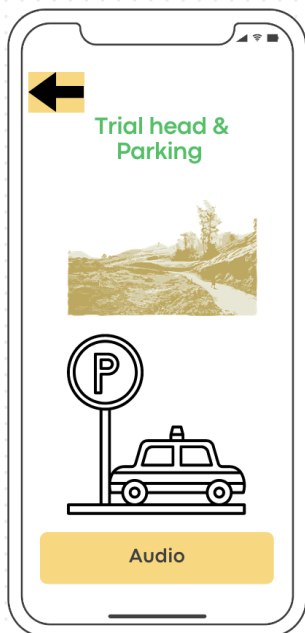
Trail screen:

On the trail screen there will be the name of the trail on the top of the page. Then different information about the trail such as the milage, rating, accessibility, and checklist. The trail description, which includes an audio recording to assist those with visual impairments in visualizing the path, is located in the lower left corner of the screen. You may access this screen by clicking the parking and trailhead button on the bottom right of the screen. The back button will return you to the screen with the list of trails at the top left of the screen. Then on the top right of the screen it has a text-to-speak button that will read all of the information on the screen.



Parking & Trailhead screen:

The parking stop and trail head photographs, as well as information on where to park for the trail, are displayed on the parking and trailhead screen. On the bottom of the screen is an audio recording giving a description of the picture up above.



3.2 Hardware Interfaces

smart phone:

The user may use the software through a mobile smart phone. Under the assumption that the user is using a traditional smartphone environment, the user may interact using touch.

3.3 Software Interfaces

- SQL Server
 - The SQL tables will contain all the information about the login information. Then another database will contain the pictures. It will be accessible by the software by Node.js, Express JS, and React Native.
- Apple IOS
- Android OS
 - Both Apple IOS and Android OS will interact with our our mobile interface to display the information to the user via a app

3.4 Communications Interfaces

The communication will be through a network using an encrypted format. The user will be accessing the mobile app interface HTTPS when accessing our software using a smart phone. The user will interact with our system through our application.

4. System Features

4.1 Map

4.1.1 Description and Priority

Description: This feature allows users to download maps of hiking trails and areas for offline use. Users can select specific regions or trails they plan to explore, and the app will download and store the corresponding map data on their device. While offline, users can access these maps, view their location, and navigate trails without requiring an internet connection.

- Priority: High
- Priority Component Ratings:
 - Benefit (8/9): This feature significantly enhances user experience by providing access to maps and navigation even in areas with poor or no internet connectivity. It improves safety and usability during hikes.
 - Penalty (5/9): Implementing map downloads may require significant development effort and ongoing maintenance, but the penalty is relatively low compared to the benefits.
 - Cost (6/9): Developing the capability to download maps for offline use and maintaining the infrastructure for map storage may incur moderate costs.
 - Risk (7/9): There is a moderate level of risk associated with this feature, including potential challenges in managing map data updates, optimizing storage, and ensuring smooth offline functionality.

4.1.2 Stimulus/Response Sequences

Use Case: Accessing Trail Maps

User Action: User opens the hiking app.

- System Response: The app's main screen is displayed.

User Action: User selects the "Find Trails" or "Explore" option.

- System Response: The app presents a list of available hiking trails.

User Action: User selects a specific hiking trail from the list.

- System Response: The app displays the trail details, including a summary, trail length, difficulty level, and elevation gain.

User Action: User taps the "View Map" button associated with the selected trail.

- System Response: The app loads the trail map for the selected trail.

User Action: Users can interact with the map by zooming in/out, panning, and exploring the trail route.

- System Response: The app provides smooth navigation and displays key trail markers, points of interest, and the user's current location if GPS is enabled.

Use Case: Downloading Trail Maps for Offline Use

User Action: User opens the hiking app.

- System Response: The app's main screen is displayed.

User Action: User selects the "Settings" or "Options" menu.

- System Response: The app opens the settings menu.

User Action: User chooses the "Offline Maps" or a similar option.

- System Response: The app presents a list of available regions or trails for offline download.

User Action: User selects a specific region or trail they want to download for offline use.

- System Response: The app initiates the download process, indicating the progress to the user.

User Action: User waits for the download to complete.

- System Response: The app notifies the user when the map is successfully downloaded and ready for offline use.

User Action: User goes to an area with poor or no internet connectivity.

- System Response: The app seamlessly switches to offline mode and allows the user to access the downloaded map and navigate the selected trail.

4.1.3 Functional Requirements

Requirement 1 (REQ-1): Display Sam Houston National Park on the Map

- Description: The system must display the geographical boundaries and features of Sam Houston National Park on the map within the hiking app.
- Response to Error Conditions or Invalid Inputs:
 - If there is a failure in loading the map data for Sam Houston National Park, the system should provide an error message to inform the user of the issue.
 - The map should gracefully handle scenarios where the map data is temporarily unavailable due to poor network connectivity, offering an option to retry loading the map.

Requirement 2 (REQ-2): Interactive Features for Sam Houston National Park

- Description: The system must allow users to interact with the map of Sam Houston National Park by enabling features such as zooming, panning, and displaying points of interest within the park.
- Response to Error Conditions or Invalid Inputs:
 - If there are errors in rendering the map features or points of interest, the system should log the error and provide a message indicating that some features may not be available at the moment.
 - The map interface should respond gracefully to user interactions, such as pinch-to-zoom or touch gestures, ensuring a smooth and intuitive experience without freezing or crashing.

4.2. Accessibility Features

4.2.1 Description and Priority

- Description: This feature focuses on enhancing the accessibility of the app to cater to users with visual impairments or mobility disabilities. It includes features such as voice-guided navigation, text-to-speech feature, and customizable font sizes. These accessibility enhancements aim to ensure that all users, regardless of their disabilities, can effectively use and navigate the app.
- Priority: High
- Priority Component Ratings:
 - Benefit (9/9): This feature provides significant benefits by making the app accessible to a wider audience, including those with visual or mobile disabilities. It promotes inclusivity and usability for all users.
 - Penalty (4/9): While implementing accessibility features may require some development effort, the penalty is relatively low compared to the considerable benefits it offers.
 - Cost (5/9): Developing and maintaining accessibility features involves moderate costs, primarily associated with design, development, and ongoing testing to ensure compliance with accessibility standards.
 - Risk (6/9): There is a moderate level of risk related to this feature, as it requires careful consideration of accessibility guidelines, potential adjustments based on user feedback, and compatibility with various assistive technologies.

4.2.2 Stimulus/Response Sequences

Use Case: Accessing Trail Information with Accessibility Features

User Action: User opens the hiking app.

- System Response: The app's main screen is displayed with voice-guided navigation and accessibility features prominently accessible.

User Action: User selects the "Find Trails" or "Explore" option.

- System Response: The app presents a list of available hiking trails with text-to-speech descriptions and customizable font sizes.

User Action: User activates the "Accessibility Options" or "Accessibility Settings" within the app.

- System Response: The app provides options to customize font sizes, color schemes, and other accessibility settings according to the user's preferences.

User Action: User selects a specific hiking trail from the list.

- System Response: The app displays comprehensive trail details, including a summary, trail length, difficulty level, elevation gain, and wheelchair accessibility status (e.g., wheelchair icon).

User Action: User taps the "View Map" button associated with the selected trail.

- System Response: The app loads the trail map with voice-guided navigation activated, offering audio directions and descriptions of key trail features.

User Action: User enables the "Text-to-Speech" feature for the trail description.

- System Response: The app converts the text-based trail information into spoken words, allowing users with visual impairments to hear the details.

User Action: User activates the "Wheelchair Accessible Trails" filter.

- System Response: The app filters the trail list to display only those trails that are wheelchair accessible, providing clear visual indicators (e.g., wheelchair icon) next to the relevant trails.

User Action: User selects a wheelchair-accessible trail.

- System Response: The app provides additional accessibility information, such as trail surface type (e.g., paved) and any potential obstacles.

4.2.3 Functional Requirements

Requirement 1 (REQ-1): Voice-Guided Navigation

- Description: The system must provide voice-guided navigation to assist users with visual impairments in exploring hiking trails. This includes providing turn-by-turn voice instructions and descriptions of trail features during navigation.
- Response to Error Conditions or Invalid Inputs:

- If the voice-guided navigation encounters difficulties in providing accurate directions or descriptions, it should notify the user and offer alternative means of navigation, such as text-based instructions.
- The system should ensure that the voice guidance feature works seamlessly with other accessibility settings, allowing users to customize the voice speed and volume to their preferences.

Requirement 2 (REQ-2): Text-to-Speech and Customizable Font Sizes

- Description: The system must incorporate a text-to-speech feature that can read aloud text-based information, such as trail summaries and descriptions. Additionally, the system must allow users to customize font sizes to suit their visual preferences.
- Response to Error Conditions or Invalid Inputs:
 - If the text-to-speech feature encounters difficulty in converting text to speech or pronounces words inaccurately, it should provide an option to read the text silently, ensuring accessibility for users with different needs.
 - When users customize font sizes, the system should immediately apply the selected font size to all relevant text elements throughout the app, ensuring a consistent and readable user interface.

4.3. Database

4.3.1 Description and Priority

- Description: This feature involves implementing a secure database system to store user login information and park photos. User login information, such as usernames and encrypted passwords, will be securely stored to ensure user account security. Additionally, the database will be used to store and organize park photos, allowing users to upload, view, and manage their photo collections within the app.
- Priority: High
- Priority Component Ratings:
 - Benefit (8/9): This feature offers substantial benefits by ensuring the security of user login information and providing a platform for users to manage their park photos. It enhances user trust and engagement.
 - Penalty (5/9): Implementing a secure database system may require significant development effort and ongoing maintenance, but the penalty is moderate compared to the benefits.
 - Cost (7/9): Developing and maintaining a secure database for user authentication and photo storage may involve moderate to high costs, including security measures, scalability, and database management.

- Risk (6/9): There is a moderate level of risk associated with this feature, primarily related to security vulnerabilities and data management. Careful planning and ongoing monitoring are essential.

4.3.2 Stimulus/Response Sequences

User Action: User installs and launches the hiking app.

- System Response: The app initializes, and the user is presented with the login screen.

User Action: User creates a new account by providing their email address and password.

- System Response: The app securely stores the user's login credentials in the database after hashing and salting the password for added security.

User Action: User logs into their existing account using their saved credentials.

- System Response: The app validates the user's login information against the stored data in the database and grants access if the credentials match.

User Action: User accesses the "My Trails" or "My Photos" section of the app to upload trail photos.

- System Response: The app allows the user to select and upload photos from their device. These photos are securely stored in the database, associated with the user's account.

User Action: User views their uploaded photos or trails in the app.

- System Response: The app retrieves and displays the user's photos and trail data from the database, ensuring a personalized experience.

User Action: User logs out of their account.

- System Response: The app ends the user's session and clears any temporary data stored locally on the device.

User Action: User logs back into their account at a later time.

- System Response: The app re-establishes the user's session and retrieves their saved data, including uploaded trail photos and personal trail information, from the database.

4.3.3 Functional Requirements

Requirement 1 (REQ-1): Secure Storage of User Login Information

- Description: The system must securely store user login information, including usernames and hashed passwords with salt, in the database.
- Response to Error Conditions or Invalid Inputs:

- If there are issues with hashing passwords or applying salt during user registration, the system should prevent the registration process, providing an error message and instructing the user to try again.
- In the event of a database connection failure during the storage of user login information, the system should log the error, notify the user of the issue, and allow them to attempt the registration or login process later.

Requirement 2 (REQ-2): Storage and Retrieval of Trail Photos

- Description: The system must have the capability to securely store and retrieve trail photos uploaded by users, associating each photo with the respective trail.
- Response to Error Conditions or Invalid Inputs:
 - If there are errors in uploading or storing trail photos due to issues like file format incompatibility, the system should inform the user of the problem and provide guidelines on acceptable photo formats and sizes.
 - When retrieving trail photos, if the system encounters data corruption or any issues with the stored photos, it should log the error and notify the user that the requested photo is temporarily unavailable.

4.4 Trail Overview

4.4.1 Description and Priority

- Description: This feature involves enhancing the app by providing comprehensive trail summaries for each hiking trail. Users will be able to access detailed information about each trail, including trail difficulty, length, elevation gain, notable features, and user reviews. This feature aims to offer users a valuable resource for choosing trails that match their preferences and skill levels.
- Priority: High
- Priority Component Ratings:
 - Benefit (8/9): This feature offers substantial benefits by improving the user experience and helping hikers make informed decisions. It enhances user engagement and satisfaction.
 - Penalty (5/9): Implementing trail summaries may require significant development effort to gather and present trail data, but the penalty is moderate compared to the benefits.
 - Cost (6/9): Developing and maintaining the infrastructure for trail summaries may involve moderate costs, including data collection and presentation.

- Risk (4/9): There is a relatively low level of risk associated with this feature, primarily related to data accuracy and user satisfaction. Thorough testing and data validation can mitigate these risks.

4.4.2 Stimulus/Response Sequences

Use Case: Viewing Trail Summaries

User Action: User opens the hiking app.

- System Response: The app's main screen is displayed, with options to explore hiking trails.

User Action: User selects the "Find Trails" or "Explore" option.

- System Response: The app presents a list of available hiking trails.

User Action: User scrolls through the list of hiking trails and selects a specific trail.

- System Response: The app displays the summary page for the selected trail.

User Action: User reads the summary, which includes information such as trail difficulty, length, elevation gain, notable features, and user reviews.

- System Response: The app provides a concise and informative summary of the trail, making it easy for the user to assess its suitability.

User Action: User decides to explore another trail and returns to the list of available trails.

- System Response: The app navigates the user back to the list of trails, allowing them to choose a different trail to explore.

User Action: User selects a different trail from the list.

- System Response: The app displays the summary page for the newly selected trail, including all relevant details.

Use Case: Navigating Back to Trail List

User Action: User opens the hiking app.

- System Response: The app's main screen is displayed, with options to explore hiking trails.

User Action: User selects the "Find Trails" or "Explore" option.

- System Response: The app presents a list of available hiking trails.

User Action: User selects a specific trail to view its summary.

- System Response: The app displays the summary page for the selected trail.

User Action: User decides to return to the list of available trails.

- System Response: The app provides a navigation option (e.g., a "Back to Trail List" button) that allows the user to return to the list of trails.

User Action: User taps the "Back to Trail List" button.

- System Response: The app navigates the user back to the list of available trails, where they can explore other trail options.

4.4.3 Functional Requirements

Requirement 1 (REQ-1): Display Trail Summaries

- Description: The system must display a summary for each hiking trail in the app, including information such as trail difficulty, length, elevation gain, notable features, and user reviews.
- Response to Error Conditions or Invalid Inputs:
 - If there is missing or incomplete trail information, the system should display a message indicating that certain details are not available for that trail but still present the available information.
 - When a user selects a trail for which the summary data is temporarily unavailable due to network issues, the system should inform the user of the issue and suggest checking back later or refreshing the trail information.

Requirement 2 (REQ-2): Navigation to Trail Details

- Description: The system must provide users with the ability to navigate from the trail summary to more detailed information about the selected trail.
- Response to Error Conditions or Invalid Inputs:
 - If there are issues with navigating to detailed information, such as broken links or data inconsistencies, the system should inform the user of the problem and offer alternative means to access trail details, such as returning to the trail list.

4.5. Security

4.5.1 Description and Priority

- Description: This feature focuses on enhancing the security of user data by implementing robust encryption algorithms and the use of salt. User data, including login credentials and sensitive information, will be securely encrypted using industry-standard encryption techniques. Additionally, the inclusion of salt will further fortify the protection of user data from unauthorized access.
- Priority: High
- Priority Component Ratings:

- Benefit (9/9): This feature offers critical benefits by ensuring the security and privacy of user data. It enhances user trust and compliance with data protection regulations.
- Penalty (5/9): Implementing strong encryption and salting mechanisms may require significant development effort, but the penalty is moderate compared to the significant security benefits.
- Cost (6/9): Developing and maintaining secure data encryption practices may involve moderate costs, primarily associated with security audits and updates.
- Risk (8/9): There is a high level of risk associated with this feature, primarily related to data breaches and security vulnerabilities. Ensuring proper implementation and ongoing security measures is critical.

4.5.2 Stimulus/Response Sequences

Use Case: Protecting User Data with Encryption and Salt

User Action: User opens the hiking app.

- System Response: The app's main screen is displayed, with options to explore hiking trails or access their user account.

User Action: User selects the "User Account" or "Profile" option.

- System Response: The app prompts the user to log in or create a new account.

User Action: User chooses to create a new account.

- System Response: The app presents a registration form, prompting the user to enter their personal information, including a username and password.

User Action: User submits their registration information.

- System Response: The app securely stores the user's login credentials, utilizing encryption algorithms and salt to protect the sensitive data.

User Action: User logs back into their account.

- System Response: The app retrieves the user's login credentials securely from the encrypted storage and grants access to their account.

User Action: User explores hiking trails and views trail summaries within the app.

- System Response: The app provides access to trail information and summaries while ensuring the user's login data remains protected in the background.

4.5.3 Functional Requirements

Requirement 1 (REQ-1): User Data Encryption and Protection

- Description: The system must encrypt and protect user data, including login credentials and sensitive information, using strong encryption algorithms and unique salt for each user's data.
- Response to Error Conditions or Invalid Inputs:
 - If there are issues with encrypting user data, the system should log the error, prevent the data from being stored in an unencrypted format, and provide a generic error message indicating a temporary issue with data protection.
 - In cases where there is a failure in applying salt to user data during encryption, the system should log the error and prevent the data from being stored or accessed without proper protection, displaying an error message to the user.

Requirement 2 (REQ-2): Secure Data Retrieval and Decryption

- Description: The system must securely retrieve and decrypt user data when authorized users access their accounts or relevant information.
- Response to Error Conditions or Invalid Inputs:
 - If there are errors in retrieving or decrypting user data, the system should log the error, deny unauthorized access, and provide a user-friendly error message, guiding the user to contact support or retry the operation later if necessary.

5. Other Non-functional Requirements

5.1 Performance Requirements

- Opening the app should take less than 1 second
- Navigating between pages on the app should take less than 1 second
- Audio recordings should start 1 second after play button is pressed
- Enhancing text size should take less than 3 seconds
- Text to Speech should take less than 2 seconds to begin after button is pressed

5.2 Safety Requirements

- The trail information provided in the app should be accurate
Action that must be taken: Trail summaries and overviews to be updated regularly
- Checklist should address *recommended* equipment to bring on hike
- Trails labeled accessible to mobile and visual impairments must be accurate

5.3 Security Requirements

- The user's login information should be secure enough so that anyone except for the developers will know that user's username/email and password
- When the user types his or her password to login, it should be hidden
- The system should be secure enough to prevent the corruption of data from unauthorized users

5.4 Software Quality Attributes

5.4.1 Availability

- The system must deliver services to the client when requested
- The application should be available 99.9% of the time

5.4.2 Dependability

- The application must be reliable enough so that when different user perform the same task, the expected outcome does not differ
- The application must be reliable enough so that when users perform normal functions, the system does not fail

5.4.3 Usability

- The application must be easy enough to learn so that users know how to use the product entirely on their first try
- The application must be more efficient than similar products

5.4.4 Flexibility

- The application must be able to support Apple iOS and Android OS

- The application must be flexible enough to support user with different national forest

5.5 Business Rules

- Users with visual impairments are able to use text-to-speech, text expand feature and audio recordings of the trails
- Users that are handicapped accessible will have the ability to view trails that are beneficial to the user
- Users with visual impairments will have the ability to view trails that are accessible to the user

6. Other Requirements

- One full database for tokenism for users' accounts
- A bucket for photos of trails
- User data is protected by encryption algorithm/salt

Appendix A: To Be Determined List

- Still unsure how to go about the audio recording on how to store them
- The with the API on when to tell the server that we want it to be offline to not try to connect with it when the user is not on wifi or cell services